Week 10 Level 1: Introduction

Random Art and My Own Commands

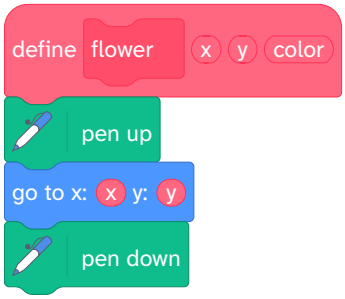
`random` makes surprises; `def` makes your own command, just like My Blocks.

What you will learn

- How to make your own command with `def`, with input slots called **parameters**.
- The difference between **defining** a command and **calling** (using) it.
- How `random` makes a new picture every time you press Run.
- How to plan your final project: **My Interactive Art Show**.

The big idea: make your own command

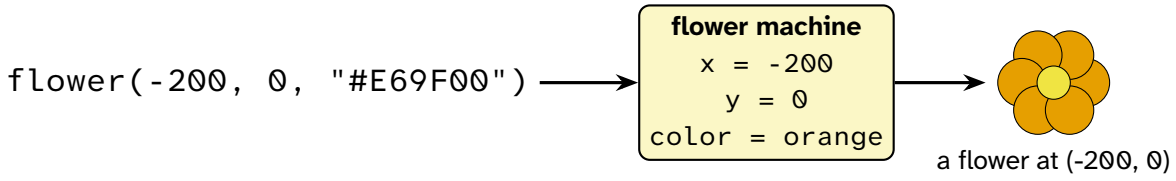
In Scratch you made My Blocks. Python has them too! You write `def`, a name, and the input slots in brackets.



→

```
def flower(x, y, color):
    t.penup()
    t.goto(x, y)
    t.pendown()
    t.color(color)
    ...
```

Defining writes the recipe. Nothing is drawn yet! **Calling** uses the recipe. The values in the brackets go into the slots, in order:



Tip: Remember

Define first (at the top), call later (at the bottom). Calls start at the left edge — not pushed in — or they become part of the recipe.

Exercise 10.1: Flower Stamp

Folder: `week10/ex-flower-stamp`. Run the starter: nothing is drawn! The recipe is there, but nobody calls it.

1. At the bottom, call it three times: `flower(-200, 0)`, `flower(0, 0)`, `flower(200, 0)`.
2. Add a third parameter: `def flower(x, y, color):` and use `t.color(color)`.
3. Give each call a color: `flower(-200, 0, PALETTE[0])`, and so on.

Done looks like: three flowers in different colors, each with a yellow middle. Then run `check_my_work.py`.

Surprise! The random module

`random.randint(1, 6)`	Like rolling a dice: a whole number from 1 to 6.
`random.randint(-350, 350)`	A random spot from left (-350) to right (350).
`random.choice(PALETTE)`	Like picking a crayon from a bag with your eyes closed.

Put `import random` at the top of your program, next to `import turtle`.

Exercise 10.2: Random Bubble Art

Folder: `week10/ex-random-art`. The starter draws 30 bubbles... all in the same place! Inside the loop, change the fixed numbers to random ones: `x`, `y`, `size` with `random.randint`, and `color` with `random.choice(PALETTE)`. Press Run three times. Is it different each time?

Your final project: My Interactive Art Show

For weeks 10 to 12 you will make your own program for an art show. It says hello, asks the viewer one or two questions, and draws a picture that changes because of the answers. It must use these **six ingredients**:

☐ a question with `input`

- ☐ a variable that remembers an answer or a number
- ☐ an `if` that chooses what to draw
- ☐ a `for` loop that repeats something
- ☐ `random` for a surprise
- ☐ a function you made with `def`

Ideas: a fish party under the sea, a garden of flowers, a city skyline at night, space with planets, a birthday card, a pattern machine. Write three ideas you like:

Then fill in the **plan sheet** in your Project Brief (Milestone M1): a sketch, your questions, and your function's name and inputs.

Tip: Be an original artist

Borrowing an idea is fine when you make it your own and say thank you. Write a comment like `# Flower idea remixed from Exercise 10.1`. Questions in your project ask about choices — colors, numbers, day or night — never about who someone is.

Vocabulary box

Function A command you make yourself, with a name. Like a My Block.

def The Python word that starts a function's recipe: `def flower(x, y, color):`.

Parameter An input slot of a function, like `x`, `y` or `color`.

Call Using a function: `flower(0, 0, "#CC79A7")`.

random A Python toolbox for surprises: `randint` and `choice`.

Remix Making something new from someone's idea, and giving them credit.

Reflection

What command would you like to invent for your project? What would its input slots be?

Why do you think a function is useful when you need ten flowers?

Week 10 Level 1: Introduction

My Art Show Plan

15–20 minutes. This week's homework is your project plan (Milestone M1).

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Finish your plan sheet (about 10 minutes)

Use the **plan sheet** in your Project Brief (page 4). Make sure it has:

- ☐ a **sketch** of your picture, with the parts labeled;
- ☐ your **question(s)**: 1 or 2 questions about choices (a number, a color, day or night);
- ☐ your **function**: its name and its input slots, like `fish(x, y, color)`;
- ☐ a few words for each ingredient: my loop repeats..., my `if` decides..., my random surprise is..., my variable remembers...

Bring the plan sheet next week. You will build from it!

★★ Challenge (want more?): Write your function on paper

Write the recipe for your function. Start with the line below, then add the steps (penup, goto, pendown, then the drawing). If you have a computer, try it in your project file.

```
def _____(x, y, _____):
```

Creative (make it yours): Colors and a title

Choose 3 colors from the color-blind-safe palette for your picture and give your artwork a title.

orange #E69F00	blue #0072B2
sky blue #56B4E9	vermillion #D55E00
green #009E73	purple #CC79A7
yellow #F0E442	black #000000

My 3 colors: _____

The title of my artwork: _____

Self-check: tick each box when it is true

- ☐ My sketch shows what my program will draw.
- ☐ My questions are about choices, not about who someone is.
- ☐ My function has a name that says what it draws.
- ☐ I know where each of the six ingredients will go.